

Excel Structure — Detailed Lecture Notes

JKSSB Computer Awareness

October 8, 2026

24.1 What Excel is and what a workbook is

Microsoft Excel is a spreadsheet program. A spreadsheet is a grid of boxes in which you can store text, numbers, dates, and formulas, and in which the program can calculate results automatically. Excel is one of the standard office applications, and its saved document is called a **workbook**.

A **workbook** is the Excel *file* — the single item that you open, edit, and save with a file name. When you start Excel and see a grid, you are looking at one **worksheet** that belongs to a workbook. A workbook can contain many worksheets, and each worksheet is a separate page of the same file.

Definition (Workbook). *The Excel file itself. A single workbook can hold one or more worksheets and is saved under one file name.*

Definition (Worksheet). *A single sheet of rows and columns inside a workbook. It is also called a sheet, and it has a tab at the bottom of the Excel window.*

At the bottom of the Excel window you will see one or more **sheet tabs** (for example, Sheet1, Sheet2, Sheet3). A new classic workbook opens with three worksheets, and you can add or remove sheets. Clicking a tab brings that worksheet to the front. The key point for the exam is the relationship: the workbook is the container (the file), and the worksheet is the content (one page inside it).

24.2 Workbook vs worksheet

These two words are a classic near-neighbour pair, so keep the contrast sharp. The workbook is the file; the worksheet is a part of the file. One workbook can contain many worksheets, but a worksheet never contains a workbook. A worksheet is not a separate saved file — it lives inside the workbook and is saved as part of it.

Point	Workbook	Worksheet
What it is	The Excel file	One sheet inside a workbook
How many	One file	Many per workbook
Saved as	A single file name	Part of that file
Tab	—	A tab at the bottom

Example 24.2.1. *An item states, “A worksheet is the Excel file, and a workbook is a part of it.” This is wrong. The relation is the other way round: the workbook is the file, and the worksheet is one page inside it. The wrong option has swapped the two terms.*

24.3 The worksheet grid: rows and columns

Every worksheet is a large grid made of **rows** and **columns**. Rows run horizontally across the sheet and are identified by **numbers**, which appear down the left edge (1, 2, 3, ...). Columns run vertically down the sheet and are identified by **letters**, which appear across the top (A, B, C, ...).

The column labels do not stop at Z. After A–Z, the labels continue as two-letter names: AA, AB, ..., AZ, then BA, BB, ..., BZ, and so on. This scheme continues with three-letter names until the final column. In Excel 2007 and later, a worksheet has:

Property	Value in Excel 2007 and later
Total rows	1,048,576 (numbered 1 to 1,048,576)
Total columns	16,384 (labelled A to XFD)
First column	A
Last column	XFD

These two figures are worth memorising because they are common direct-recall items. In older Excel versions (Excel 97–2003, the `.xls` format) the limits were much smaller: 65,536 rows and 256 columns, with the last column labelled IV. Modern Excel uses the larger grid ending at column XFD.

24.4 The cell and the active cell

A **cell** is the small rectangular box formed where a row and a column meet. It is the basic unit of a worksheet, and it is where you actually type your data. The whole sheet is just a very large collection of cells.

At any moment, exactly one cell is the **active cell**. The active cell is the cell that is currently selected: it has a thick border around it, and anything you type is entered into it. The active cell's address is shown in the Name Box, and its contents are shown in the Formula Bar. Even when you select a group of cells, only one of them — the active cell — is the one the next typed entry goes into.

Definition (Cell). *The rectangular box formed by the intersection of a row and a column, in which data is entered.*

Definition (Active cell). *The cell that is currently selected. It has a thick border, receives the next typed entry, and has its address shown in the Name Box.*

24.5 Cell addresses and references

Every cell has a unique **cell address** (also called a cell reference). The address is written with the **column letter first**, followed by the **row number**. For example:

- **B5** means column B, row 5.
- **C7** means column C, row 7.
- **D12** means column D, row 12.

The order is fixed: letter then number. Writing 12D, B:5, or B_5 is not a valid Excel address. Because column labels can have more than one letter, an address such as AB100 is also valid: it means column AB, row 100.

24.6 Ranges

A **range** is a group of two or more cells treated as one block. It is written by typing the addresses of two diagonally opposite corners and separating them with a **colon** (:). For example, **A1:C10** means all the cells from column A to column C and from row 1 to row 10. A range is normally rectangular, and the cells in it are contiguous.

The number of cells in a range is the number of columns multiplied by the number of rows. For example, the range B2:D5 covers three columns (B, C, D) and four rows (2, 3, 4, 5), so it contains $3 \times 4 = 12$ cells.

Definition (Range). *A rectangular group of two or more selected cells, written as two corner addresses joined by a colon, such as A1:C10.*

Example 24.6.1. *A question lists B5 and calls it a range. This is wrong: B5 is a single cell, because it has only one address and no colon. A range must have two corner addresses joined by a colon, such as B5:D9. The colon is the signal that a range, not a single cell, is meant.*

24.7 The formula bar

The **Formula Bar** is the long white bar that sits below the Ribbon and above the worksheet grid. It always shows the *contents* of the active cell. If the active cell holds plain text or a number, the Formula Bar shows that text or number. If the active cell holds a formula, the Formula Bar shows the *formula itself* (for example =A1+A2), while the cell itself shows the calculated result.

You can also edit the active cell in the Formula Bar by clicking in it. This is useful when a formula is long. Remember the division of labour: the cell displays the result, and the Formula Bar displays the underlying formula or data.

Definition (Formula Bar). *The bar below the Ribbon that shows, and allows editing of, the contents of the active cell — including the formula behind a calculated result.*

24.8 The name box

The **Name Box** is the small box at the *left end of the Formula Bar*. It normally shows the **address of the active cell** (for example, B5). If a range of cells is selected, it shows the name or the address range of that selection (for example, A1:C10). You can also type an address into the Name Box and press Enter to jump directly to that cell.

Definition (Name Box). *The box at the left of the Formula Bar that shows the address of the active cell or the name of the selected range.*

Keep the two bars apart. The Name Box (on the left) answers “which cell am I in?” The Formula Bar (the long part) answers “what is in that cell?”

24.9 Basic data types

Excel accepts several basic kinds of entry in a cell:

- **Text** (also called a label): letters, words, names, and headings. Text is not used in arithmetic.
- **Number**: a numeric value that can be used in calculations.

- **Date and time:** entered and displayed as a date or time, but stored internally by Excel as a number.
- **Formula:** an entry that begins with the = sign and calculates a result from other cells.

Excel also recognises **logical values** (TRUE and FALSE) and displays **error values** such as #VALUE! when a calculation cannot be completed. By default, Excel aligns these data types in a helpful way:

Data type	Default alignment	Example
Text	Left	Roll No
Number	Right	4500
Date	Right	15-08-2026

A quick visual check therefore tells you what Excel thinks the entry is: a value sitting on the *left* is treated as text, and a value sitting on the *right* is treated as a number (or a date, which is stored as a number). This default alignment is a common exam point.

24.10 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. A **workbook** is the file; a **worksheet** is one page inside it. Do not swap them.
2. A **cell** is one box; a **range** is two or more cells joined by a colon.
3. A cell address is **column letter first, then row number** (B5), never the reverse.
4. A worksheet in Excel 2007+ has **1,048,576 rows** and **16,384 columns**; the last column is **XFD**, not IV (IV was the old .xls limit).
5. The **Name Box** shows the address; the **Formula Bar** shows the contents.
6. Only **one cell is active** at a time, even when a range is selected.

Example 24.10.1. *An item asks, “What is the total number of rows in a modern Excel worksheet?” Recall the pair: rows are the larger count, 1,048,576, and columns are 16,384. If the stem instead asks for the last column, the answer is XFD; if it asks for the number of cells in a range, multiply columns by rows.*

Exercises

1. Distinguish a workbook from a worksheet.
2. What is a cell? How is a cell address written, and give one example.
3. How many rows and how many columns does a worksheet have in Excel 2007 and later, and what is the last column label?
4. What is a range? Write the range that covers columns A to C and rows 1 to 10, and state how many cells it contains.

5. State what the Name Box and the Formula Bar each display.
6. List four basic data types that can be entered in an Excel cell and give the default alignment of text and of numbers.

Solutions

1. A workbook is the Excel file itself, saved under one file name; a worksheet is a single sheet of rows and columns inside a workbook. A workbook can contain many worksheets.
2. A cell is the box formed where a row and a column intersect. A cell address is written as the column letter followed by the row number, for example B5 (column B, row 5).
3. A worksheet has 1,048,576 rows and 16,384 columns; the last column is labelled XFD.
4. A range is a rectangular group of two or more cells written as two corner addresses joined by a colon. The range is A1:C10, and it contains $3 \times 10 = 30$ cells.
5. The Name Box shows the address of the active cell (or the name of the selected range). The Formula Bar shows the contents of the active cell, including the formula behind a calculated result.
6. Four basic data types are text, number, date/time, and formula. Text is aligned to the left by default, and numbers to the right by default.